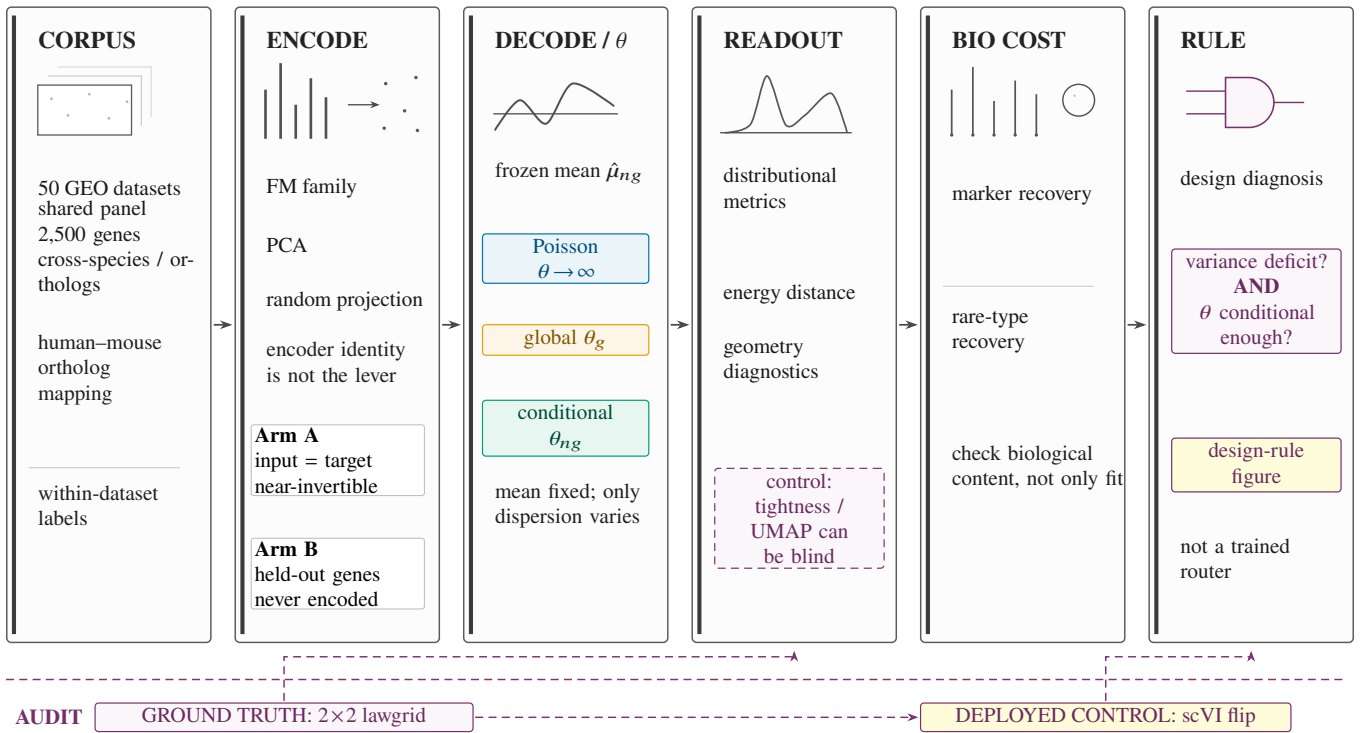

CORPUS
50 GEO datasets
shared panel
2,500 genes
cross-species / orthologs
human–mouse ortholog mapping
within-dataset labels
ENCODE
FM family
PCA
random projection
encoder identity is not the lever
Arm A
input = target
near-invertible
Arm B
held-out genes
never encoded
DECODE / θ
frozen mean $\hat{\mu}_{ng}$
Poisson
$\theta \to \infty$
global θ_g
conditional
θ_{ng}
mean fixed; only dispersion varies
READOUT
distributional metrics
energy distance
geometry diagnostics
control: tightness / UMAP can be blind
BIO COST
marker recovery
rare-type recovery
check biological content, not only fit
RULE
design diagnosis
variance deficit?
AND
θ conditional enough?
design-rule figure
not a trained router
AUDIT
GROUND TRUTH: 2×2 lawgrid
DEPLOYED CONTROL: scVI flip